

PES1UG24CS613

SANJITH KS

Screenshots and analysis

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ make test-integration

--- Third Commit ---
Committed: ba556d6ce1f5... Add farewell

--- Full History ---
commit ba556d6ce1f5dec60109423d3a6631fc23c42df6135a87f6ba97ae1da8eb8201
Author: sanjith PES1UG24CS613
Date: 1776321028

    Add farewell

commit cd56a6367b08a8f356514c92b56574361faeaf7dd37fb3fd7cefae42f3f0d178
Author: sanjith PES1UG24CS613
Date: 1776321028

    Update file.txt

commit 04831b8e1014961843bae7e63ae5c4c15743acd121688b07fc027d3ae42aa3c4
Author: sanjith PES1UG24CS613
Date: 1776321028

    Initial committial commitPESQ

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
ba556d6ce1f5dec60109423d3a6631fc23c42df6135a87f6ba97ae1da8eb8201

--- Object Store ---
objects created:
10
.pes/objects/04/831b8e1014961843bae7e63ae5c4c15743acd121688b07fc027d3ae42aa3c4
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/ba/556d6ce1f5dec60109423d3a6631fc23c42df6135a87f6ba97ae1da8eb8201
.pes/objects/cd/56a6367b08a8f356514c92b56574361faeaf7dd37fb3fd7cefae42f3f0d178
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafaaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff

=== All integration tests completed ===
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$
```

1a)

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ make clean
make test_objects
/test_objects
|
~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check
```

2)

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$
```

Tree.c

2a

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$
```

2b

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ xxd .pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6 | head -20
00000000: 7472 6565 2039 3600 3130 3036 3434 2062  tree 96.100644 b
00000010: 7965 2e74 7874 00e6 7ed6 6bcb 4a70 8250  ye.txt...~.k.Jp.P
00000020: a89f 315d 4d8b b927 0334 3c84 df83 4438  ...lJM...'4<...D8
00000030: 880e 13a6 8652 bc31 3030 3634 3420 6865  ....R.100644 he
00000040: 6c6c 6f2e 7478 7400 05c8 d777 d0be 8574  llo.txt...w...t
00000050: 8b67 80d1 b5ca f7d6 9754 e6e6 8060 c265  .g.....T...`e
00000060: ca21 cbd6 c05e 5f43  .1...^_C
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$
```

Index.c

3a

```
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ ./pes add file2.txt object.c index.c
./pes status
Staged changes:
  staged:   file1.txt
  staged:   file2.txt
  staged:   index.c
  staged:   object.c

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked: tree.c
  untracked: test_objects.c
  untracked: pes.c
  untracked: index.h
  untracked: pes.h
  untracked: Makefile
  untracked: README.md
  untracked: test_sequence.sh
  untracked: commit.c
  untracked: test_tree.c
  untracked: tree.h
  untracked: commit.h
  untracked: .gitignore
```

```

luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776319331 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776318860 6 file2.txt
100644 9163276a9e4bafc2316d88db142baf9cfd473f5491ef6ba06d4a086bbd070b6 1776319304 9246 index.c
100644 141c7e89b84227bea8a0fc3789c708b03185d00fe7a97fd34494c10b7fb9d07f 1776315642 9135 object.c
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$

```

Commit.c

4a)

```

gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Committed: 1d4a8bfecc5d... Initial commit
Committed: c38d0ac864c5... Add world
Committed: 125fac9c8f08... Add farewell
commit 125fac9c8f080416fbadbbcac1786b8675e475cbbc745a28adf7da167a412a84
Author: sanjith PES1UG24CS613
Date: 1776320094

    Add farewell

commit c38d0ac864c56942186094a2751e588249fdf1ecff6205aca8d257c18ac393ba
Author: sanjith PES1UG24CS613
Date: 1776320094

    Add world

commit 1d4a8bfecc5d901c5941dc3e413ce21b0ac24c711038713249205ea29245412b
Author: sanjith PES1UG24CS613
Date: 1776320094

    Initial committial commit

luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$

```

4b

```

luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/12/5fac9c8f080416fbadbbcac1786b8675e475cbbc745a28adf7da167a412a84
.pes/objects/1d/4a8bfecc5d901c5941dc3e413ce21b0ac24c711038713249205ea29245412b
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/c3/8d0ac864c56942186094a2751e588249fdf1ecff6205aca8d257c18ac393ba
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$

```

4c

```
• luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$ cat .pes/refs/heads/main  
cat .pes/HEAD  
125fac9c8f080416fbadbbcac1786b8675e475cbbc745a28adf7da167a412a84  
ref: refs/heads/main  
○ luffy@sanjith:~/os_u4_orange/PES1UG24CS613-pes-vcs$
```

Analysis Questions

Q5.1

A branch is just a file in `.pes/refs/heads/`. For `pes checkout <branch>`:

- Read the commit hash from `.pes/refs/heads/<branch>`
- Change `.pes/HEAD` to:
 - `ref: refs/heads/<branch>`
- Read that commit's tree and update the working directory to match it
- Rewrite `.pes/index` to match the checked out files

This is difficult because files may exist in one branch but not another, and local changes should not be overwritten.

Q5.2

Before checkout, compare every tracked file with the index:

- Compare current file size and mtime with the values in `.pes/index`
- If they changed, recompute the file hash
- If the file is modified locally and also different in the target branch, refuse checkout

Example:

Current branch: README = A

Target branch: README = B

Working directory: README = C

Since $C \neq A$ and $B \neq A$, checkout should stop.

Q5.3

Detached HEAD means .pes/HEAD stores a commit hash directly instead of a branch name.

You can still make commits, but no branch points to them. If you switch away, those commits become unreachable.

They can be recovered by creating a new branch that points to the commit hash.

Q6.1

To find unreachable objects:

- Start from every branch in .pes/refs/heads/
- Follow commit → tree → blob links
- Store every visited hash in a hash set
- Delete any object in .pes/objects/ that is not in the set

A hash set is useful because lookup is fast.

For 100,000 commits and 50 branches, garbage collection may need to visit a few hundred thousand objects.

Q6.2

GC should not run while a commit is being created.

Example:

1. A new blob is written
2. The commit has not yet been written
3. GC runs and sees the blob is unreachable
4. GC deletes it
5. The commit is written and now points to a missing blob

Git avoids this using lock files and by not deleting recently created objects immediately.